

Public Economics and Development

Week 5: Theories on Provision of Public Goods

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Roadmap for today

1. Market failure and the role of the government

2. Provision of public goods and welfare

2.1 Optimal provision of public goods

2.2 Effects of providing welfare

3. Management of public services in reality

3.1 Liebman, Mahoney (2017): Do Expiring Budgets Lead to Wasteful Year-End Spending? Evidence from Federal Procurement

3.2 Rasul, Rogger (2018): Management of Bureaucrats and Public Service Delivery: Evidence from Nigerian Civil Service

Where does the money raised by government go to

Previous lecture: How to set tax system to optimally collect money

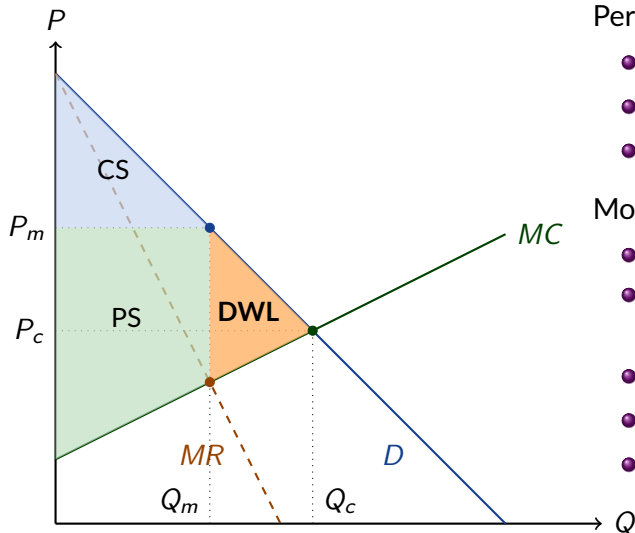
- Maximizing social welfare
- With respect to meeting the revenue requirement or budget constraint
- Often comes with efficiency loss *when compared to perfectly competitive outcomes*

Why do government want to raise money? Correct for market failures

- Markets may not be competitive - monopolies and oligopolies
- There are incomplete information - asymmetric information
- Markets under/over produce some goods (e.g. public goods) - externalities
- Markets may generate inequitable outcomes - redistribution

We focus on public goods & redistribution, and discuss management of public institutions

Imperfect competition



Perfect competition

- Individual firms as price taker
- $P = MR = MC$
- Equilibrium at (Q_c, P_c)

Monopolies and social welfare

- Firm = market
- $P > MR = MC$
 - ▶ Lower unit price for additional goods
- Equilibrium at (Q_m, P_m)
- Lower consumption at higher prices
- Antitrust law, public ownership

Hidden type problem (adverse selection)

Example: Health insurance market

- Private information about type $\theta \in \{\theta_L, \theta_H\}$ exists *before* contracting
- Insurer observes only the population distribution
- Pooled premium P^* attracts only high-risk types
 - ▶ Low-risk types find P^* too high relative to their expected benefits
 - ▶ Insurers' expected payouts exceed P^* and market unravels
- Separating types of contracts also costly
 - ▶ To deter high-risk types from mimicking, low-risk contract offers incomplete coverage
 - ▶ Incomplete insurance for low-risk types
- **State:** Eliminate this selection issue through national health care system

Asymmetric information — hidden action (moral hazard)

Example: Health insurance market

- After contracting, the insured no longer bears the full cost of their actions
- The effective price of medical care to individuals falls from P_0 to cP_0 ($0 < c < 1$)
- Reduced costs leads to overconsumption or risky behaviors
 - ▶ Increased careless behaviors
 - ▶ Overconsumption of medical services
- Problem arises due to high costs of monitoring actions post-contract
- Optimal insurance contract balances between risk-sharing and incentives
 - ▶ First-best requires full information
 - ▶ Inevitable trade-off between coverage and distortion in incentives

Both asymmetrical information problems can be applied to labor markets, taxation, welfare programs, and public service delivery

Externalities

Definition

- An activity that affects another entity directly
- Without cost or benefits to the one conducting these activities
- Negative externality hurts others without the cost to the actor
- Positive externality benefits others without the benefit to the actor

Examples

- Environmental Pollution: Pollutors do not usually bear costs
- Vaccination: Others do not pay you to get vaccinated
- *Public goods*: Private entities are not compensated for social benefits

We treat this in more detail in this lecture

Redistribution: Income support for the poor

Income support programs are key methods of redistribution in many countries

- Unemployment insurance
- Earned income tax credit
- Universal basic income
- Negative income tax

Active debates on redistribution

- How much welfare support is desirable?
 - ▶ Important source of income for poor households
 - ▶ Labor disincentives for non-poor (tax) and poor (phase-out rate per additional income)
- Selective targeting (means-tested, application-based) vs universal application (e.g. UBI)
- What type of support: Cash vs in-kind

Provision of public goods and welfare

Definition and examples of a public good

A pure public good is non-excludable and non-rivalrous in consumption

- Nonexcludable: Once provided, no consumer can be excluded from consumption
- Nonrivalrous: Consumption by one does not reduce the availability for others

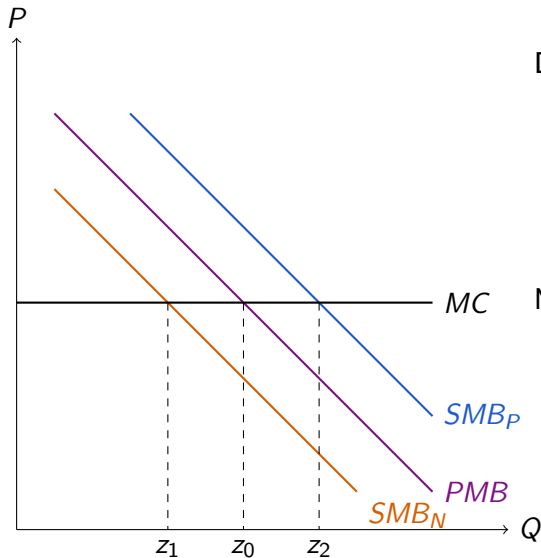
In practice, it is difficult to satisfy both properties

- Many publicly available goods and services can be depleted: Common goods
- Some goods do not deplete in availability but are limited-access: Club goods

Why public goods are different?

- Nonexcludability → each consumer rely on others to provide the public good
- This leads to *free-rider problem* and underprovision of these goods
- Classic positive externality problem

Graphical illustration of externality



Definition

- Externality: When cost or benefit on 3rd party is not reflected in market prices
- Social marginal benefit (SMB)
 $\neq$ Private marginal benefit (PMB)

Market vs socially optimal outcome

- Market: $PMB = MC \rightarrow z_0$
- Positive externality: $SMB_P = MC \rightarrow z_1$
 - ▶ Markets underprovide the good
- Negative externality: $SMB_N = MC \rightarrow z_2$
 - ▶ Markets overprovide the good

Mathematical analysis of externalities

Setup with two goods and two persons

- Utility for both (good 2 (z_i) induces externality)

$$U_1 = x_1 + u_1(z_1) + v_1(z_2)$$

$$U_2 = x_2 + u_2(z_2) + v_2(z_1)$$

where if $v_i' > 0 \rightarrow$ positive externality and otherwise negative

- Individual and social budget constraints are as follows

$$\text{Individual (IBC): } x_i + pz_i \leq \omega_i \quad (i \in \{1, 2\})$$

$$\text{Social (SBC): } x_1 + x_2 + pz_1 + pz_2 \leq \omega_1 + \omega_2$$

Mathematical analysis of externalities

Market outcomes

- Each individual maximizes own utility subject to IBC

$$\max_{x_i, z_i} x_i + u_i(z_i) + v_i(z_j) + \lambda[\omega_i - x_i - pz_i]$$

- FOC w.r.t $[x_i]$: $\lambda = 1$
 - FOC w.r.t $[z_i]$: $u'_i(z_i) - \lambda p = 0 \Rightarrow u'_i(z_i) = p$
- Each individual equates their own marginal benefits with marginal costs (price of z)

Mathematical analysis of externalities

Optimal outcomes

- From society's perspective, it maximizes (where SWF = sum of individual utility)

$$\max_{x_1, x_2, z_1, z_2} x_1 + x_2 + u_1(z_1) + u_2(z_2) + v_1(z_2) + v_2(z_1) + \mu[\omega_1 + \omega_2 - x_1 - x_2 - pz_1 - pz_2]$$

- ▶ FOC w.r.t $[x_i]$: $\mu = 1$ for any i
- ▶ FOC w.r.t $[z_1]$: $u'_1(z_1) + v'_2(z_2) - \mu p = 0 \Rightarrow u'_1(z_1) + v'_2(z_2) = p$
- ▶ FOC w.r.t $[z_2]$: $u'_2(z_2) + v'_1(z_1) - \mu p = 0 \Rightarrow u'_2(z_2) + v'_1(z_1) = p$
- In positive externality ($v'_i(z_j) > 0$: SMB_P)
 - ▶ SMB_P is above PMB : Too little consumption of z compared to desired levels
- In negative externality ($v'_i(z_j) < 0$: SMB_N)
 - ▶ SMB_N is below PMB : Too much consumption of z compared to desired levels

Possible solutions through internalization

Pigouvian taxation and internalization

- Note that socially optimal z_1 satisfies

$$u'_1(z_1) + v'_2(z_1) = p$$

- Person 1 will internalize person 2's benefit from z_1 if price charged to person 1 (q_1) is

$$u'_1(z_1) = q_1 = p - v'_2(z_1)$$

- Effectively charge tax (or provide subsidy) t_1 with

$$t_1 = q_1 - p = -v'_2(z_1)$$

Note that if $v'_2(z_1) < 0$ then $t_1 > 0$ (tax), otherwise $t_1 < 0$ (subsidy)

- May need individualized prices $\rightarrow$ difficult to implement
- Other solutions: Mandate quantity through quantity controls (govt needs perfect info)

Back to public goods

Socially optimal provision of public goods

- Social utility comes from consumption of private goods x and public good z

$$SWF : W(U_1(x_1, z), U_2(x_2, z)) = U_1(x_1, z) + U_2(x_2, z)$$

- To characterize feasible allocation, we use the production possibilities frontier

$$PPF : F(x_1, x_2, z) = 0, \Rightarrow x_1 + x_2 + z \leq E$$

- Social optimum is derived from following Lagrangian

$$\mathcal{L} = U_1(x_1, z) + U_2(x_2, z) + \lambda[E - x_1 - x_2 - z]$$

Back to public goods

Solutions

- FOC w.r.t. x_i ($i = 1, 2$): $[x_i] : U_{i,x_i} - \lambda = 0 \implies U_{i,x_i} = \lambda$
- FOC w.r.t. z : $[z] : U_{1,z} + U_{2,z} - \lambda = 0 \implies (U_{1,z} + U_{2,z}) = \lambda$
- Since $\lambda = U_{1,x_1} = U_{2,x_2}$ from $[x_i]$, substitute into $[z]$:

$$U_{1,z} + U_{2,z} = U_{1,x_1} = U_{2,x_2}$$

- Dividing through by λ and using $U_{1,x_1} = U_{2,x_2} = \lambda$:

$$\frac{U_{1,z}}{U_{1,x_1}} + \frac{U_{2,z}}{U_{2,x_2}} = 1$$

- Since $MRS_i^{zx} = U_{i,z}/U_{i,x_i}$ and $MRT = 1$ (linear PPF):

$$\sum_i MRS_i^{zx} = MRT^{zx}$$

Optimal provision of public goods

Samuelson Condition

$$\sum_i MRS_i^{zx} = MRT^{zx}$$

- Sum of individual value of z in terms of x = Cost of producing z in terms x
- This contrasts with efficient private good consumption (call it y), where

$$MRS_i^{yx} = MRT^{yx}$$

i.e. person i need not consider person j

- Efficient allocation requires knowledge on true value of public goods to *each* person
 - ▶ This information is private: People can understate their value to free-ride
 - ▶ This is a preference revelation problem (markets fail to induce people to be truthful)
 - ▶ This shows why private provision of public goods is difficult

Income maintenance program: Mathematical setup

Individual's problem

- Maximize its utility w.r.t. budget constraint (labor earnings)

$$\max_{X,L} u(X, L) \quad \text{s.t.} \quad X \leq w(T - L)$$

- ▶ X : Consumption
- ▶ L : Leisure (time not spent working)
- ▶ T : Maximal time per day
- ▶ w : Wages (per hour)
- ▶ $T - L$: Labor (I will use l interchangeably)

Assumptions

- $u_X > 0, u_L > 0, u_{XX} < 0, u_{LL} < 0$ (X and L : Normal goods, diminishing marginal returns)

Income maintenance program: Mathematical setup

Income maintenance program

- Ensures individuals earn at least G through this benefit formula

$$B = \max[G - tE, 0]$$

- ▶ E : Labor earnings ($w(T - L) = wI$)
- ▶ t : Tax rate on additional earnings

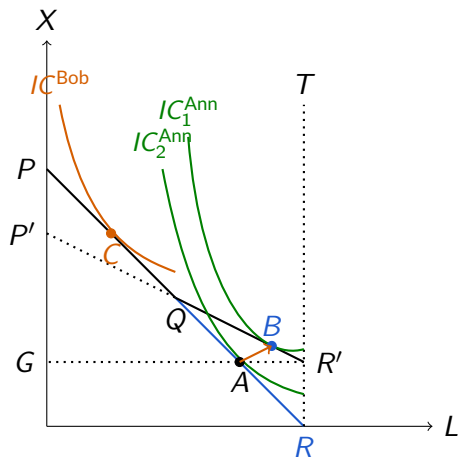
Revised individual problem

- Budget constraint: $X \leq w(T - L) + B$, which can be written as

$$X \leq \max[(1 - t)w(T - L) + G, w(T - L)]$$

- Practical approach: Solve Two lagrangian (for each BC) $\rightarrow$ Choose highest Utility

Income maintenance program and labor supply



Visualizing income maintenance program

- Pre-program budget constraint: PQR
- Post-program budget constraint: PQR'

Ann's choice

- Moves from A to B
- Consumption and leisure $\uparrow$, labor $\downarrow$
- Effect of t on labor: Substitution dominates
 - ▶ Substitution effect: labor $\downarrow$
 - ▶ Income effect: labor $\uparrow$

Bob's choice

- Choice unaffected at C

Explaining labor disincentives

How substitution effect works

- Recall the budget constraint

$$X \leq \max[(1 - t)w(T - L) + G, w(T - L)] = \max[(1 - t)wl + G, wl]$$

- Without t : For additional unit of labor l , get w
- With t : Additional unit of labor only returns $(1 - t)w$
- Larger $t \rightarrow$ larger substitution effect: Labor results in less budget (and consumption)
- Explains why many beneficiaries opt to not work

But these do not apply to everyone

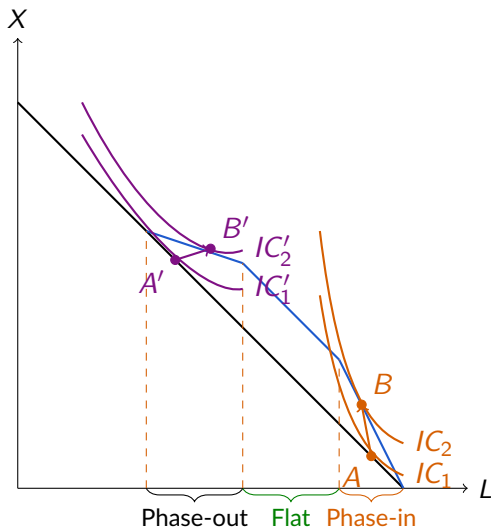
- If $E > \frac{G}{t}$, budget constraint unaffected ($\because \max[G - tE, 0] = 0$, like Bob)
- t may raise labor through income effect
 - Higher $t \rightarrow$ Less income (poorer) $\rightarrow$ Less leisure (and subsequently more labor)
 - Overall effect of t depends on relative strength of substitution and income effects

Different types of income support program: Earned income tax credit

Design of EITC setup

- Refundable tax credit for low-to-moderate income workers in the US
- EITC is conditional on earned income — explicitly designed to reward work
- Benefit formula has three regions (based on labor income y)
 - ▶ **Phase-in** ($y < G$): Credit rises with y at rate τ_s until G – subsidize labor supply
 - ▶ **Flat** ($G \leq y \leq G'$): Maximum credit $\tau_s \cdot G$ maintained over a y between G and G'
 - ▶ **Phase-out** ($G' < y < G''$): Credit withdrawn at rate τ_r for y beyond G' , phased out at G''
- Mathematic formula
 - ▶ **Phase-in:** $w(T - L) + \tau_s w(T - L)$
 - ▶ **Flat:** $w(T - L) + \tau_s G$
 - ▶ **Phase-out:** $(1 - \tau_r)w(T - L) + \tau_s G - \tau_r G'$
- Encourage labour force participation while providing income support

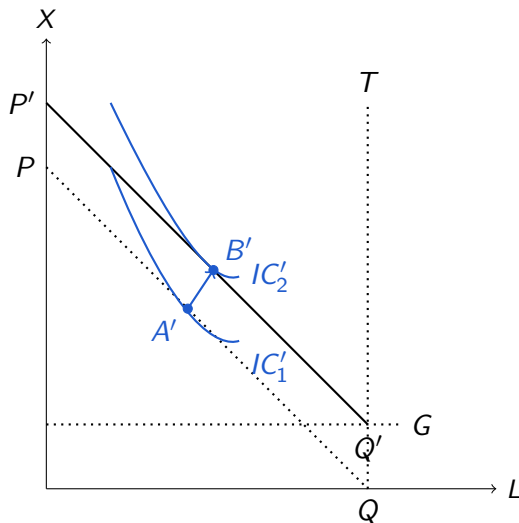
EITC: Budget constraint and labor supply



Labor supply effects by region

- **Phase-in:** effective wage $(1 + \tau_s)w > w$
 - ▶ Substitution effect: $l \uparrow$
 - ▶ Income effect: $l \downarrow$ slightly
- **Flat:** lump-sum transfer, effective wage $= w$
 - ▶ Income effect: $l \downarrow$ slightly
 - ▶ No substitution effect
- **Phase-out:** effective wage $(1 - \tau_r)w < w$
 - ▶ Substitution and income effect: $l \downarrow$

Different types of income support program: Universal basic income



Changes in the constraints

- Unconditional transfer G to every individual
- Budget constraint simply shifts up by G
- Effective wage unchanged $\Rightarrow$ *no substitution effect*, only income effect
- Income effect predicts reduction in labor supply (leisure is a normal good)

Practical concerns

- Fiscal cost: universal coverage of G requires high tax rates elsewhere
- Targeting: Transfers to non-poor households

Different types of income support program: Negative income tax

Theoretical motivations behind all above income maintenance programs

- Friedman (1962): government sets a guaranteed income G and a tax rate t on earnings

$$T(E) = tE - G$$

negative when $E < G/t$ (transfer), positive when $E > G/t$ (tax)

- The income maintenance program is a NIT: $B = \max[G - tE, 0]$
- Special cases:
 - ▶ EITC: adds a phase-in region where $t < 0$ (subsidy on earnings)
 - ▶ UBI: sets $t = 0$ – pure lump-sum G , no substitution effect, only income effect
- All three share the same equity–efficiency trade-off
 - ▶ Higher G improves redistribution but requires higher t to finance
 - ▶ Higher t distorts labor supply through substitution effect

Different types of income support program: Unemployment insurance

Individuals pay a premium while working and receive benefits when unemployed

- Suppose that α percent of population is employed ($1 - \alpha$ unemployed)
 - ▶ Assume that everyone has asset A and have same utility function $u(\cdot)$
 - ▶ If employed: Labor income w with disutility d , pay premium T

$$X_e \leq w - T + A$$

- ▶ If unemployed: No labor income, but receive benefit B

$$X_u \leq A + B$$

- Government: Finances the program with taxes from employed, thus the constraint

$$\alpha T = (1 - \alpha)B$$

- Need to prevent employed population 'mimicking' unemployed

$$u(\text{employed}) \geq u(\text{unemployed})$$

Constraints in solving the social welfare problem for UI

Mathematical setup of the social welfare problem

$$\max_{T,B} \alpha[u(w - T + A) - x] + [(1 - \alpha)u(A + B)] \quad \text{s.t.} \quad \begin{cases} \alpha T = (1 - \alpha)B & \text{(BC)} \\ u(w - T + A) - x \geq u(A + B) & \text{(IC)} \end{cases}$$

- We plug in the BC to write the optimization as

$$\begin{aligned} \max_B \alpha \left[u \left(w - \frac{1 - \alpha}{\alpha} B + A \right) - x \right] + [(1 - \alpha)u(A + B)] \\ \text{s.t. } u \left(w - \frac{1 - \alpha}{\alpha} B + A \right) - x \geq u(A + B) \end{aligned}$$

Outcome and implications of UI

If we ignore the IC (incentive compatibility) constraint

- First order condition w.r.t B

$$\begin{aligned}[B] : & -\alpha \frac{1-\alpha}{\alpha} u_b \left(w - \frac{1-\alpha}{\alpha} B + A \right) + (1-\alpha) u_b(A+B) = 0 \\ \iff & w - \frac{1-\alpha}{\alpha} B + A = A + B \\ \iff & \alpha w = B\end{aligned}$$

- When this is added to individual budget constraints, consumption is equalized at

$$X_u = A + \alpha w, \quad X_e = w - (1-\alpha)W + A = A + \alpha w \Rightarrow X_u = X_e$$

- Full insurance, but no incentives for anyone to work in this case

Outcome and implications of UI

If we consider the IC (incentive compatibility) constraint

$$\alpha \left[u \left(w - \frac{1-\alpha}{\alpha} B + A \right) - x \right] + [(1-\alpha)u(A+B)] + \lambda \left[u \left(w - \frac{1-\alpha}{\alpha} B + A \right) - x - u(A+B) \right]$$

- First order condition w.r.t λ : IC becomes binding and $\lambda > 0$ (Kuhn-Tucker condition)

$$[\lambda] : u(w - T + A) - x = u(A + B)$$

- First order condition w.r.t B :

$$[B] : -u_B \left(w - \frac{1-\alpha}{\alpha} B + A \right) + u_B(A+B) = \lambda \left[\frac{u_B(w - \frac{1-\alpha}{\alpha} B + A)}{\alpha} + \frac{u_B(A+B)}{1-\alpha} \right] > 0$$
$$\Rightarrow A+B < A - \frac{1-\alpha}{\alpha} B + w \text{ (or } X_u < X_e) \Rightarrow B < \alpha w$$

- Incentive compatible, but partial insurance $\rightarrow$ possible mimicking leads to second-best

Takeaways from public goods provision

Governments can correct for market failures

- Can achieve socially optimal allocation of public goods
- Also provide redistribution (through different mechanisms)

However, even some socially optimal outcome are unachievable by government

- Governments also suffer from asymmetrical information problems
- For solutions that require private information, government may not do better
 - ▶ Any deviation from perfect competition → first-best solution are unachievable in any case
- We also assume that government can implement any plans
 - ▶ Governments do create inefficient spending
 - ▶ The bureaucratic personnel and administrative backdrop may not be capable

Liebman, Mahoney (2017):
Do Expiring Budgets Lead to Wasteful Year-End
Spending? Evidence from Federal Procurement

Q: Do organizations spend wastefully at end of the year?

Decision structure at organizations

- Budgets expire at end of fiscal year (remainder are returned)
- Organizations rush to spend that money, even at low-value projects
 - ▶ Low opportunity costs: These funds are unusable next year
- Liquidity-constrained through out the year: Hold back expenditure early, spend late

This paper examines the cause and consequences of year-end spending

- Uses US federal government procurement data
- Documents increasing late-year spending (vs other parts of the year)
- Drop-off in quality of projects spent on at late-year
- Policy suggestion: Allow for rollover to next year (some departments do)

Simple model

Budgeting model for the government agency

- An agency makes the spending (x_m) decision for each half-year ($m \in \{1, 2\}$)
- The spending generates $\alpha_m v(x_m)$ value (α_m : stochastic, v is concave)
- The funds are allocated to the agency by the congress
- Agency solves

$$\max_{x_1, x_2} \alpha_1 v(x_1) + E_{\alpha_2}[\alpha_2 v(x_2)] \quad \text{s.t.} \quad x_1 + x_2 \leq B$$

- Congress solves at beginning of each year

$$\max_B E_{\alpha_1, \alpha_2}[\alpha_1 v(x_1) + \alpha_2 v(x_2) - \lambda(x_1^* + x_2^*)]$$

- Due to concavity of v and uncertainty about $m = 2$ at planning, $x_2^* > x_1^*$
- Due to high x_2 and concavity of v , quality of spending ($\alpha_m v(x_m)/x_m$) is lower in $m = 2$

Data sources and empirical strategy

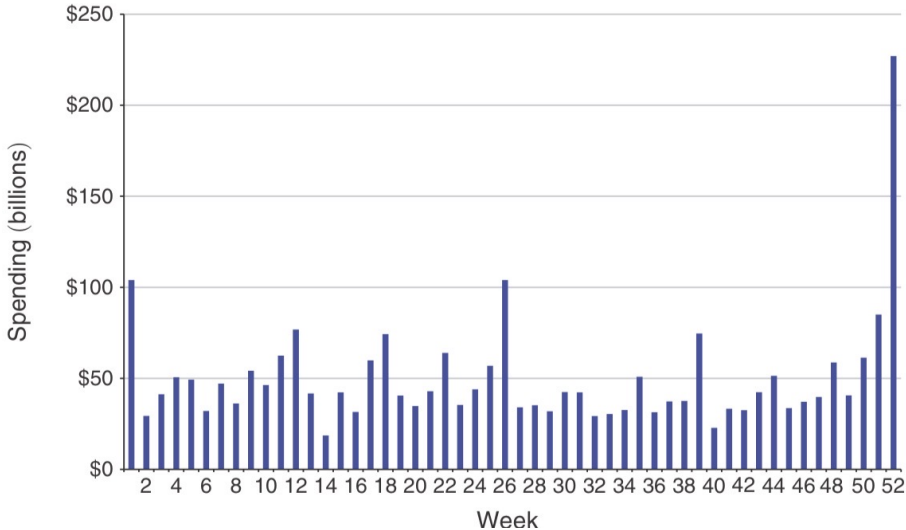
Data from federal procurement data (2004-2009)

- USAspending.gov is the primary source (for all contracts)
- Total value of contract, and breakdown of costs are available
- Available across multiple government branches (for quality analysis, rely on IT projects)

Regression analysis

- Quality analysis: Ordered logit regression
 - ▶ Odds ratio as outcome: Odds ratio $1/2$ = chances of higher categorical value is 50% lower
 - ▶ Conversely, it means odds of lower category are 2 times as great
- Effect of rollovers: Department of justice (with rollover) vs others around 1992
 - ▶ DOJ authorized to rollover 4% of annual revenue in 1992 for IT projects
 - ▶ Compare 1994-2006 period spending

Spending bunched at the end of fiscal year



Chances of low quality investment: 2.3-5.6 times higher at last week

TABLE 4—ORDERED LOGIT REGRESSIONS OF OVERALL RATING ON LAST WEEK
AND CONTROLS

	Odds ratio of higher overall rating			
	(1)	(2)	(3)	(4)
Last week	0.26 (0.07)	0.43 (0.12)	0.28 (0.09)	0.18 (0.07)
Year fixed effects		X	X	X
Agency fixed effects			X	X
Project characteristics fixed effects				X
Observations	671	671	671	671

Notes: Odds ratios from ordered logit regressions. Coefficient of 1 indicates no effect. Overall rating is a quality index that combines the CIO evaluation, cost rating, and scheduling rating subindices (see text for details). It takes values from 0 to 10, with 10 being the best. Project characteristics are fixed effects for investment phase, service group, and line of business (see online Appendix Table A4). Observations weighted by inflation-adjusted spending. Robust standard errors in parentheses.

Source: IT Dashboard data, accessed March 2010, <http://it.usaspending.gov>

- Inverse of 0.18 and 0.43 are 5.6 and 2.3, respectively

Rollover reduces year-end spending by 9.5 percentage points

TABLE 6—DIFFERENCE-IN-DIFFERENCES OF LAST WEEK SPENDING ON JUSTICE AND IT

	Non-IT	IT	Difference	Difference-in-differences
Other agencies	0.085 (0.004) [N = 13,589,187]	0.120 (0.015) [N = 558,587]	−0.036 (0.017)	
Justice	0.093 [N = 418,187]	0.033 [N = 2,192]	0.060	−0.095 (0.017)

Notes: Table shows fraction of spending in the last week of the year at the Department of Justice and other agencies on non-IT and IT contracts. Standard errors clustered by agency in parentheses, calculated using the corresponding regressions specifications. Standard errors for the DOJ-level statistics are undefined and are thus omitted from the table.

Source: Federal Procurement Data System, accessed October 2010, www.usaspending.gov

Rasul, Rogger (2018):
Management of Bureaucrats and Public Service
Delivery: Evidence from Nigerian Civil Service

Q: Do management practices matter for public organizations?

Government effectiveness matters, but what contributes to it?

- Economic analysis in public sector focused on selection and incentives
- Not much on managerial practices applied to middle-tier of bureaucrats

This study estimates correlation between public service and management practices

- Background: Nigeria (Largest population in Africa, has inefficient practices)
- Data collected through survey on management practices and outputs of agencies
- Better autonomy increases completion rates (incentive/monitoring does the opposite)
 - ▶ Incentives are misaligned,
 - ▶ Bureaucrats spend more time gathering performance indicators vs actually working

Data sources and empirical strategy

Primary datasets

- Details on 4700 public sector projects (since 2006) handcoded (8% of social spending)
- Above data are from civil society members and independent engineers
- Management practice from surveys on Nigerian public sector workers
 - ▶ Focus on autonomy (have say in input, flexibility) vs incentives and monitoring

Research design exploits variation in practices across organizations doing similar projects

$$y_{ijn} = \gamma_1 \text{Autonomy}_n + \gamma_2 \text{Inc/Mon}_n + \gamma_3 \text{other}_n + \beta X_{ijn} + \lambda_j + e_{ijn}$$

- i : project, j : project type, n : organization
- e.g. Small dams constructed by three different organizations
- Outcome: Project completion rates (note: 30+% of projects not even started in NGA)
- Does not fully control for endogenous selection of projects
 - ▶ Still have non-random differences across autonomy vs incentive/monitoring practices

Project completion correlated with management practices

Management Practices and Public Sector Service Delivery

Standard errors: clustered by project type within organisation
OLS estimates

	(1) Unconditional	(2) Organisation controls	(3) Project controls	(4) Project type fixed effects	(5) Binary project initiation	(6) Quality-adjusted completion rate
CS-autonomy	0.11** (0.05)	0.18*** (0.03)	0.17*** (0.03)	0.18*** (0.03)	0.15*** (0.03)	0.11** (0.04)
CS-incentives/monitoring	-0.06* (0.03)	-0.11*** (0.02)	-0.11*** (0.02)	-0.14*** (0.02)	-0.16*** (0.02)	-0.08*** (0.02)
CS-other	0.10*** (0.04)	0.05 (0.03)	0.05 (0.03)	0.08*** (0.02)	0.06** (0.03)	0.08*** (0.02)
Organisation controls (capital, general, noise)	No	Yes	Yes	Yes	Yes	Yes
Project controls	No	No	Yes	Yes	Yes	Yes
Fixed effects	None	None	None	Project type	Project type	Project type
Adjusted R ²	0.07	0.29	0.29	0.32	0.36	0.17
Observations (clusters)	4,721 (201)	4,721 (201)	4,721 (201)	4,721 (201)	4,721 (201)	2,206 (144)

Why might incentive-based practices be ineffective?

Heterogeneity analysis (Table 5 in the paper)

- Negative effects exacerbated for more complex projects
 - ▶ These require more varied effort types to be exerted
 - ▶ Difficult to target and design optimal incentives schemes (all else equal)
- Older bureaucrats respond to incentives with influence activities (vs. actual work)
- Also worse if we have subjective (not objective) performance evaluation
- Suggestive evidence of roles played by corruption

Incentives geared towards engaging with politicians vs actual work

Bureaucrat's Engagement with Politicians and Management Practices

	Robust standard errors OLS estimates							
	Engagement with national parliamentary politicians		Engagement with state parliamentary politicians		Engagement with Local Government Chairmen		Engagement with Community/Religious Groups	
	(1a) Managers	(1b) Non managers	(2a) Managers	(2b) Non managers	(3a) Managers	(3b) Non managers	(4a) Managers	(4b) Non managers
CS-autonomy	0.19 (1.05)	0.34 (0.89)	0.04 (0.68)	0.52 (0.76)	1.98* (1.10)	1.49 (0.96)	1.08 (1.36)	-0.35 (1.17)
CS-incentives/monitoring	3.49** (1.39)	-2.46** (1.19)	3.61*** (1.06)	-0.43 (0.89)	3.79*** (1.44)	-0.32 (1.30)	3.68** (1.67)	-1.67 (1.53)
CS-other	1.03 (1.09)	1.06 (0.95)	-0.17 (0.73)	0.34 (0.88)	0.19 (1.20)	1.14 (1.22)	0.79 (1.51)	2.84** (1.41)
Organisation controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bureaucrat controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects (bureaucrat grade)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	0.08	0.04	0.02	0.02	0.07	0.06	0.06	0.06
Observations	1,175	1,184	1,130	1,122	1,140	1,131	1,192	1,202

Next class

Moving onto the developing country contexts

- Opportunity to observe Classical theoretical format more critically
- Identify what can still carry over
- More importantly, examine what *does not* carry over to developing countries

Key topics (for the remainder of this course)

- Fiscal capacity in developing countries
- Anti-poverty programs in developing countries
- Organization of public institutions